SCHOOL OF ENGINEERING

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STRENGTH OF MATERIALS LABORATORY MANUAL

NAME : ....

REGISTER NO : ....

SEMESTER : ....

ACADEMIC YEAR : ....

CONTENTS

Experiment no: Date:

1. TENSION TEST ON MILD STEEL SPECIMEN

AIM

To conduct tension test on a mild steel rod and to determine its elastic properties.

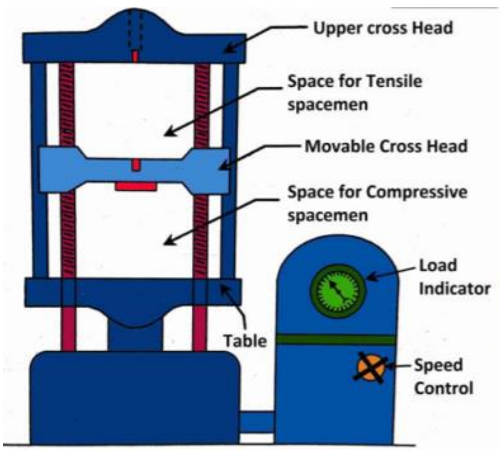
PRINCIPLE

Tension test is the basic test most widely used to determine various mechanical properties of ductile materials. In this test, the test specimen is subjected to axial tensile force in a UTM till failure. An elastically deformed solid will return to its original form as soon as load is removed. However, if the load is too large, the material can be deformed permanently. The initial part of the tension curve which is recoverable immediately after unloading is termed as elastic and the rest of the curve which represents the manner in which solid undergoes plastic deformation is termed plastic. The stress below which the deformations essentially entirely elastic is known as the yield strength of material. During plastic deformation, at larger extensions strain hardening cannot compensate for the decrease in section and thus the load passes through a maximum and then begins to decrease. This stage the “ultimate strength”’ which is defined as the ratio of the load on the specimen to original cross sectional area, reaches a maximum value. Further loading will eventually cause ‘neck’ formation and rupture. During the course of the load application, the test specimen will behave differently (elastic, elastic-plastic, plastic deformation) and various mechanical properties can be calculated from the observations made during the loading stage.

APPARATUS:

1. Universal Testing Machine (UTM)
2. Mild steel specimens
3. Graph paper
4. Scale
5. Vernier caliper or micrometer

DIAGRAM: UNIVERSAL TESTING MACHINE



PROCEDURE

1. Measure the mean diameter (D_m) of the specimen with a micrometer or vernier caliper by taking readings at three different places along the length of the specimen.
2. Calculate the mean area of cross section of the specimen (A_m).
3. Calculate the gauge length. Round off the obtained value to the nearest 5mm accuracy (L_i).
4. Calculate the approximate load at yield point (P_y approx) and ultimate load (P_u approx) by assuming the yield and ultimate stresses as 250 MPa and 400 MPa respectively.
5. Mark a line at 50 mm (grip length) from either ends of the specimen for effective gripping, in the UTM.
6. Mark the middle of the specimen with a centre punch. Select a convenient interval (s) by dividing the gauge length with a suitable number. Mark on either sides of the midpoint of the specimen at the selected interval (s) till the marking for the grip is reached.
7. Select the load range of the measuring unit based on the approximate load calculated.
8. Fix the specimen between the upper and middle cross heads of the UTM in such a way that the markings for the grips are in line with the edge of grips.
9. Adjust the gauge length of the extensometer to the calculated value (Li).
10. Fix the extensometer firmly at the middle of the test specimen.
11. Close both the control valve and return valve of the loading unit. Switch on the machine and apply load on the specimen without any jerk by gradually opening the control valve. Apply the load to about one third of the calculated approximate yield point load (P_y approx.), stop loading by closing the control valve, switch off the machine and gradually reduce the load to zero by opening the return valve. Repeat this procedure three times.
12. Set the loading and dummy pointer of the measuring unit to zero. Set the dial gauges of the extensometer to zero.
13. Load the specimen gradually at an approximate strain rate of 0.01 per minute as explained in step 11. At a constant interval of load, observe the dial gauge reading of extensometer. Continue this process of observing the load and dial gauge readings until the load reaches one third of the calculated approximate yield point load (P_y approx) or about 5 readings are taken and remove the extensometer from the specimen.
14. Continue the loading and carefully observe the load indicating pointer trajectory for identifying yield point load (P_y). When yield point reaches, the needle may stop for a while or may jerk with continued deformation of the specimen.
15. Continue the loading further with a reduced rate of loading and observe the scale formation and its pattern on the specimen.
16. Once the ultimate load is reached, neck will form on the specimen. The load indicator will start returning back and the dummy pointer will stay at the maximum load reached (P_u). Reduce the rate of loading and carefully observe the load indicating pointer trajectory and note down the load corresponding to breaking of the specimen (P_b).
17. Remove the specimen from the grips, observe the fracture surfaces. Join the two pieces together and measure the neck diameter and the elongated length (L_f) corresponding to the initial gauge length (L_i). Ensure that the fracture surface is almost at the middle of the L_i.

18. Plot a stress vs. strain graph from the calculated values and determine, the modulus of elasticity of the material of the specimen by determining the slope of the initial straight portion of the curve.
19. Compare the results obtained with the standard values and write inference.

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of vernier caliper	
2	Least count of extensometer	
3	Range of measuring unit	
4	Least count of measuring unit	

Mean diameter of rod (D_m)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter = MSR+(VSR x LC) (mm)	Mean diameter (D_m) mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

SL NO	PARTICULARS	VALUE
1	Mean area of cross section, $A_m = \frac{\pi Dm^2}{4}$	
2	Theoretical value of gauge length = $5.65\sqrt{A_m}$	
3	Gauge length of the specimen	
4	Approximate yield point load (P_y approx)=250 x A_m	
5	Approximate ultimate load (P_u approx) =400 x A_m	

Load extensometer observation table

SL NO	Load P_i (kN)	Extensometer Reading			Elongation ΔL (mm)	Strain $\epsilon_i = \frac{\Delta L}{L_i}$	Stress $f_i = \frac{1000 \times P}{A_m}$ (MPa)
		Left	Right	Mean			
1							
2							
3							
4							
5							
6							
7							

SAMPLE CALCULATION

Serial number of the observation –

Elongation, ΔL = (Mean extensometer reading) x (Least count of extensometer)

=

Strain $\epsilon_I = \frac{\Delta L}{L_i}$ =

Stress $f_i = \frac{1000 \times P}{A_m}$ =

(INSERT GRAPH)

SL NO	PARTICULARS	VALUE
1	Load at yield point (P_y)	
2	Ultimate load (P_u)	
3	Breaking load (P_b)	
4	Increased length corresponding to gauge length L_f	
5	Diameter of the specimen at the neck D_f	
6	Area of cross section of the specimen at the neck, A_f $=\frac{\pi Df^2}{4}$	

CALCULATIONS

Young's modulus (from graph), E =

Yield stress f_y $= \frac{1000 \times P_y}{A_m} =$

Ultimate stress f_u $= \frac{1000 \times P_u}{A_m} =$

Nominal breaking stress $f_b = \frac{1000 \times P_b}{A_m} =$

Actual breaking stress $f_{bactual} = \frac{1000 \times P_b}{A_f} =$

Percentage elongation$= \frac{(Lf - Li) \times 100}{Li} =$

Percentage reduction in area $= \frac{(Af - A_m) \times 100}{A_m} =$

Maximum shear stress $\boldsymbol{\tau}_{max} = \frac{f_u}{2} =$

RESULTS

SL NO	PARTICULARS	VALUE
1	Young's modulus (from graph), E	
2	Yield stress f_y	
3	Ultimate stress f_u	
4	Nominal breaking stress f_b	
5	Actual breaking stress $f_{bactual}$	
6	Percentage elongation	
7	Percentage reduction in area	
8	Maximum shear stress	

INFERENCE

SL NO	PARTICULARS	STANDARD VALUE	OBTAINED VALUE
1	Young's modulus E		
2	Yield stress f_y		
3	Ultimate stress f_u		
4	Percentage elongation		
5	Percentage reduction in area		
6	Maximum shear stress		

Experiment no: **Date:**

2. TORSION TEST ON MILD STEEL SPECIMEN

AIM

To conduct torsion test on mild steel specimen to find modulus of rigidity or to find angle of twist of the materials.

PRINCIPLE

When a uniform prismatic circular bar is subjected to a torque every section perpendicular to the axis of the bar will be subjected to a state of pure shear. The moment of resistance developed by the shear stress will be equal in magnitude and opposite in sense to the applied force. So the torsion formula is used to determine various elastic properties of the material of the specimen.

Torsion equation:

$$\frac{T}{J} = \frac{C\theta}{L} = \frac{\tau}{R}$$

T= maximum twisting torque (N mm)

J= Polar moment of inertia (mm^4)

τ = shear stress N/mm^2

C = modulus of rigidity N/mm^2

θ = angle of twist in radians

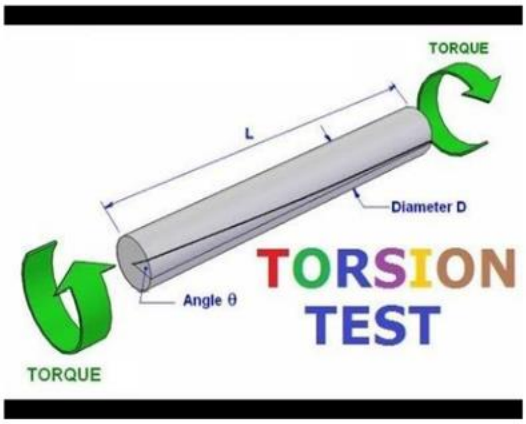
L = length of shaft under torsion (mm)

Along with the transverse shear stress, complementary shear stresses will also develop in the direction of the axis of the torsion specimen. So if the shear strength of a material is less than its tensile or compressive strength, then the material will fail by transverse shear when subjected to torsion. In the other hand, if the shear strength of a material is more than its tensile or compressive strength, then the material will fail either in tension or in compression.

APPARATUS:

1. A torsion test machine along with angle of twist measuring attachment
2. Standard specimen of mild steel
3. Graph paper
4. Steel rule
5. Vernier caliper or a micrometer

DIAGRAM:



TORSION TESTING MACHINE

PROCEDURE

1. Measure the mean diameter (D_m) of the specimen with a vernier caliper or a micrometer by taking readings at three different places along the length of the specimen.
2. Mark 400 mm (L), symmetric to the mid length of the specimen.
3. Mount the specimen in the grip in such a way that the length of specimen between the grip is L. Tighten the grips properly by adjusting sliding jaw to avoid any slip during the loading.
4. Choose the appropriate loading range depending upon specimen.
5. Apply torque on the specimen without any jerk by gradually rotating the handle on the loading end of the specimen. After reaching about 5 kg-m torque, release the torque to zero by rotating the handle in the reverse direction. Repeat this procedure three times to ensure an elastic behaviour of the specimen.
6. Set the load pointer and the adjustable disc to zero. Note down the initial reading of the discs attached to each grip, which indicates the angle of twist.
7. Apply the torque gradually as stated in "step 5" and for every increment of 1 kg-m torque, observe the disc readings. Continue the loading till a torque of 5 to 6 kg-m torque is reached and then stop loading. Bring down the torque on the specimen to zero and remove the specimen from the grips.
8. Plot the torque – angle of twist graph from the calculated values and determine, the ratio $\frac{T}{\theta}$ by determining the slope of the initial straight portion of the curve.
9. Calculate the modulus of rigidity C by using the torsion equation.
10. Compare the results obtained with the standard values and write inference.

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of Vernier caliper	
2	Least count of discs	
3	Least count of measuring unit	

Mean diameter of rod (D_m)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter = MSR+(VSR x LC) (mm)	Mean diameter (D_m) mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

Torque –angle of twist observation table

SL NO	Applied Torque T_i (kg m)	Angle of twist (degree)			Angle of twist (θ) (radian)	Torque T (Nm)
		Left	Right	Difference (θ_i)		
1						
2						
3						
4						
5						
6						
7						
8						

SAMPLE CALCULATION

Serial number of the observation –

Angle of twist (θ) = θ_i x $(\frac{\pi}{180})$ =

Torque T (Nm) = T_i x 9.81=

(INSERT GRAPH)

CALCULATIONS

$\frac{T}{\theta}$ from graph =

Mean radius, r $= \frac{D_m}{2} =$

Polar moment of inertia J = $\frac{\pi \, Dm^4}{32}$

Modulus of rigidity, C = 1000 x $\frac{T}{\theta}$ x $\frac{L}{J}$ =

Applied torque for the calculation of shear stress, T_a = ……………...................Nm

Maximum shear stress developed corresponding to a torque of ……………….. Nm,

$\tau_{max} = 1000 \text{ x } \frac{Ta \times r}{J} =$

RESULTS

SL NO	PARTICULARS	VALUE
1	Modulus of rigidity, C	
2	Maximum shear stress developed corresponding to a torque of ……………….. Nm	

INFERENCE

Experiment no: **Date:**

3. DEFLECTION TEST ON STEEL I BEAM

AIM

To determine the modulus of elasticity of steel by performing bending test on a steel I beam.

PRINCIPLE

For any materials within elastic limit, the ratio of normal stress to normal strain is defined modulus of elasticity. A direct method of finding out E is by conducting a uniaxial tension test. An alternative procedure is to carry out a bending test on a standard section. In a bending test the test specimen is subjected to transverse loading so as to produce transverse flexural stresses on the beam. This results in shortening of fibres above the natural axis due to the compressive stress and expansion of bottom fibres due to tensile stress. Defected beam will return to initial straight configuration at least in the initial stages of loading (within the elastic limit) in such a case, as per simple bending theory, the normal stress and strain across any cross section of the beam is linear. The ratio of stress to strain within elastic limits of a material is known as young's modulus of elasticity and is given by the formula

$$\frac{M}{I} = \frac{f}{y} = \frac{E}{R}$$

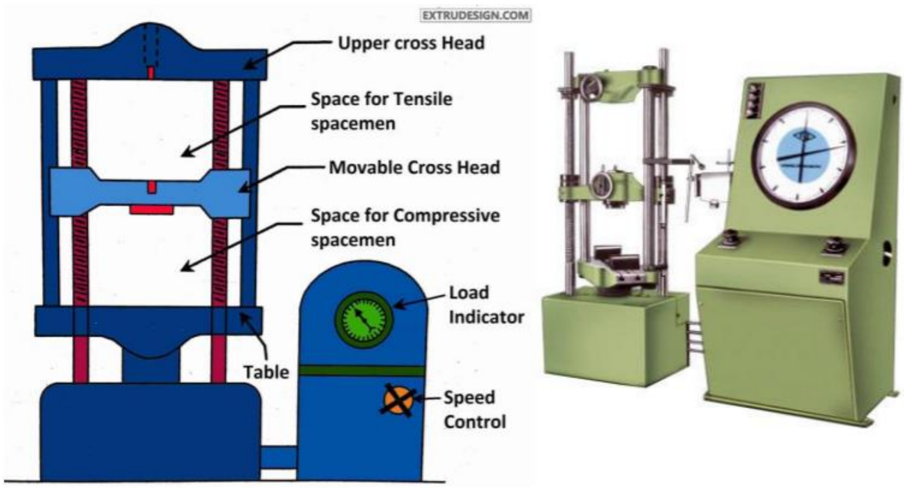
M = Bending moment
I = moment of inertia of the section
f = bending stress developed
y = distance of any fibre measured from the neutral axis.
E = modulus of elasticity
R = radius of curvature of the beam

It may be noted that, the above said formula is truly valid for a pure bending condition. The shear forces at section will also contribute to the total deflection of the structure. However, the contribution of shear in the case of deep beams will be significant whereas for beams where span to depth ratio is large, the effect due to shear is negligible.

APPARATUS:

1. Universal testing machine
2. Steel I beam specimen
3. Dial gauge
4. Graph paper
5. Steel rule
6. Vernier caliper or a micrometer

DIAGRAM:



UNIVERSAL TESTING MACHINE

PROCEDURE

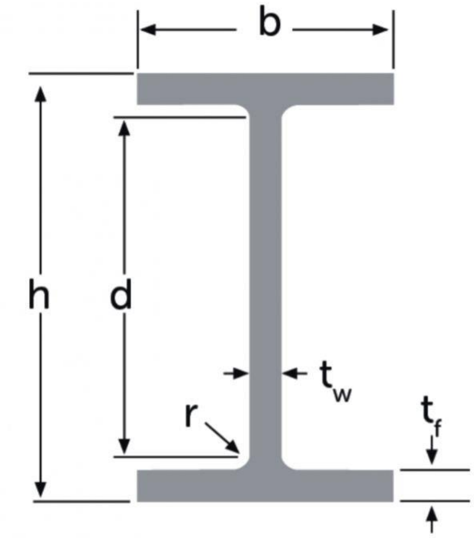
1. Measure the dimensions of the cross section of the given steel I section and compute the moment of inertia and modulus of the section.
2. Mark the mid span of the given beam and then the span length with equal spacing to either side of the central mark.
3. Calculate the range of loading for the test assuming yield strength of 165N/mm^2 and hence finalise the interval of load for which the deflection is to be measured.
4. Arrange the dial gauge (s) so as to measure the defection of the beam at the mid span.
5. Apply the load gradually at the mid span at rete of 5kN per second and note the dial gauge reading corresponding to the predetermined load interval.
6. Plot the graph with load on the Y- axis and deflection on the X axis and ascertain the condition of loading within the elastic limit.
7. Note down the slope of the graph and hence calculate the modulus of elasticity of steel using the appropriate deflection formula.

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of dial gauge	
2	Range of measuring unit	
3	Least count of measuring unit	

DIMENSION OF I SECTION



SL NO	PARTICULARS	DIMENSION (mm)			
		d_1	d_2	d_3	Mean
1	Over all Depth (h)				
2	Flange thickness (t_f)				
3	Web thickness (t_w)				
4	Depth of web (d)				
5	Breadth of flange(b)				

SAMPLE CALCULATION

Serial number of the observation –

Load =

Deflection, $\boldsymbol{\delta} =$ (Dial gauge reading) **x** (Least count of dial gauge)

Span of beam L =

Moment of inertia, I = $\frac{1}{12}$ [bh^3- (b-t_w) d^3] =

Modules of section (Z) = $\frac{I}{\frac{H}{2}}$ =

Bending Moment = f x Z

$\frac{WL}{4}$ = 165 x Z

Maximum applied load to be applied (W) = $\frac{165 \times Z \times 4}{L}$ =

Load deflection observation

SL NO	Load (kN)	Dial gauge reading (mm) (d)	Deflection δ (mm) d x least count of dial gauge
1			
2			
3			
4			
5			
6			
7			
8			

(INSERT GRAPH)

CALCULATIONS

$\frac{P}{\delta}$ from the graph =

Deflection at the mid span of simply supported, **δ** $\frac{1}{48EI}$ x P L^3

Modulus of elasticity of the material of the beam, E =$\frac{1}{48I}$ x $\frac{P}{\delta}$ x L^3 x 10^3

RESULTS

Modulus of elasticity of the material of the beam =

INFERENCE

Experiment no: **Date:**

4. SPRING TEST

(a) TEST ON CLOSED COILED HELICAL SPRING

AIM

To determine modulus of rigidity and other properties of a closed coiled helical spring.

PRINCIPLE

Helical spring is a piece of wire coiled in the form of a helix. It can undergo considerable deformation without getting permanently distorted and hence is capable of storing considerable strain energy. If the slope of the helix is small, then the spring is called as closed coiled spring· If closed coiled spring is subjected to axial forces, the wire of the spring will be subjected to pure shear. So a closed coiled spring of mean radius R with n turns and a wire diameter d is equivalent to a circular shaft of diameter d and length $L = 2\pi Rn$. If W is the axial load on the spring, the twisting moment on any section of the spring (or equivalent shaft) will be $T = WR$. Hence the general torsion formula can be used to determine various properties that are required for the design of closed coiled helical springs.

Consider a closed coiled spring of mean diameter D with n turns and a wire diameter d

Then

Spring index = D/d

Modulus of rigidity, $G = \frac{1}{\delta d^4} \text{ x } (8WD^3n)$

Stiffness of spring, $k = \frac{W}{\delta}$

Maximum shear stress in the wire corresponding to a force (W) of ……………….. kN,

$\tau_{max} = \frac{1}{\pi d^3} \text{ x } (8WD)$

Strain energy stored in the wire corresponding to τ_{max}

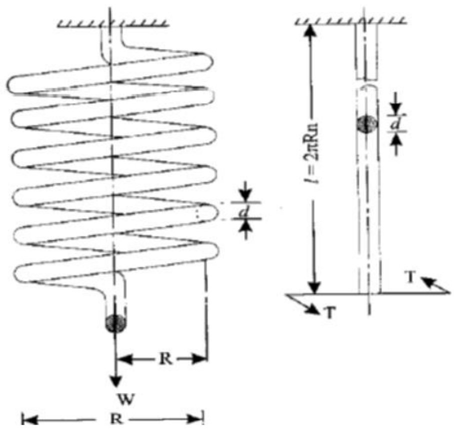
$U = \frac{1}{4G} \text{ x } (\tau_{max}{}^2 \text{ x volume of spring})$

APPARATUS:

1. Spring testing machine.
2. A closed coil spring,
3. Scale
4. Graph paper

5. Steel rule
6. Vernier caliper or a micrometer

DIAGRAM:



CLOSED COIL HELICAL SPRING **SPRING TESTING MACHINE**

PROCEDURE

1. Measure the mean diameter (d) of the wire of the spring, mean diameter of the spring (D) with a vernier caliper or a micrometer by taking readings at three different places along the length of the specimen. Also note down the number of coils (n).
2. Place the spring on hooks fixed to the upper and middle cross heads of the spring testing machine. Tighten the nuts on the top hook so that the spring is properly held in position.
3. Close the release valve of the loading unit. Apply loading by hand operation at constant rate till the load is reached about 100 kg and then release the load to zero by opening the release valve. Continue this process for three times to ensure that the spring behaves elastically.
4. Set the load pointer and deformation pointer either to zero or to a convenient value.
5. Close the release valve and apply load on the specimen without any jerk manually. At regular increments of applied load, observe the total elongation of the spring. Continue the measurement till 5 to 6 observations are made and then release the load to zero.
6. Plot the load- deflection graph from the calculated values and determine, the ratio W/$\boldsymbol{\delta}$ by determining the slope of the initial straight portion of the curve.

7. Compare the results obtained with the standard values and write inference.

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of vernier caliper	
2	Least count of loading unit	
3	Number of active coils in the spring, n	

Diameter of the spring wire (d)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter d = MSR+(VSR x LC) (mm)	Mean diameter of the spring wire (d) mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

Outer diameter of the spring (OD)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter OD = MSR+(VSR x LC) (mm)	Mean Outer diameter of the spring (OD)mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

Load – deflection observation table

SL NO	Load W_i (kg)	Load W (N)	Scale reading (mm)		Spring deformation δ= (Final – Initial) (mm)
			Initial	Final	
1					
2					
3					
4					
5					
6					
7					
8					

SAMPLE CALCULATION

Serial number of the observation –

Applied load W = W_I x 9.81=

Spring deformation, **δ** = (Final scale reading – Initial scale reading)

=

(INSERT GRAPH)

CALCULATIONS

Mean diameter of the spring, D = OD – d =

Spring index = D/d =

Stiffness of spring (from graph), k = $\frac{W}{\delta}$ =

Modulus of rigidity, G = = $\frac{1}{d^4}$ x [($\frac{W}{\delta}$) x ($8D^3n$)] =

Maximum shear stress in the wire corresponding to a force (W) of ……………….. N,

τ_{max} =$\frac{1}{\pi d^3}$ x (8WD) =

Volume of spring, V = (2πDn) x ($\frac{\pi d^2}{4}$) =

Resilience of the spring corresponding to τ_{max}

U = $\frac{1}{4G}$ x ($\tau_{max}{}^2$ x volume of spring) =

RESULTS

SL NO	PARTICULARS	VALUE
1	Spring index	
2	Stiffness of spring, k	
3	Modulus of rigidity, G	
4	Maximum shear stress in the wire corresponding to a force (W) of ……………….. N, τ_{max}	
5	Resilience of the spring corresponding to maximum shear stress of ……………………. MPa	

INFERENCE

(b) TEST ON OPEN COILED HELICAL SPRING

AIM

To determine modulus of rigidity and other properties of an open coiled helical spring.

PRINCIPLE

Helical spring is a piece of wire coiled in the form of a helix. It can undergo considerable deformation without getting permanently distorted and hence is capable of storing considerable strain energy. If the slope of the helix is appreciable, then the spring is called as an open coiled spring· If an open coiled spring of mean radius R with n turns, a wire diameter d and angle of helix α is considered, the axial force W acting on the spring, will cause a twisting moment of WR cosα and a bending moment of WR sinα on the spring wire. Accordingly, on any cross section of the spring wire, both bending and shearing stresses will develop.

Consider an open coiled spring of mean diameter D with n turns, a wire diameter d and angle of helix α. Then

Spring index = D/d

Deflection of the spring under axial load W,

$\boldsymbol{\delta} = [\frac{1}{d^4} \times (8WD^3 n \sec\alpha)] \times [(\frac{1}{G} \times \cos^2\alpha) + (\frac{1}{E} \times 2 \sin^2\alpha)]$

Stiffness of spring, $k = \frac{W}{\delta}$

Stresses in a normal section of the wire corresponding to a force W

a) Maximum shear stress due to twisting moment, $\tau = \frac{1}{\pi d^3} \times (8WD \cos\alpha)$

b) Maximum bending stress due to bending moment, $\sigma = \frac{1}{\pi d^3} \times (16WD \sin\alpha)$

Maximum shear stress in the plane of wire corresponding to a force (W)

$$\tau_{max} = \frac{1}{\pi d^3} \times (8WD)$$

APPARATUS:

1. Spring testing machine.
2. An open coiled spring,
3. Scale
4. Graph paper
5. Steel rule
6. Vernier caliper or a micrometer

DIAGRAM:

OPEN COIL HELICAL SPRING SPRING TESTING MACHINE

PROCEDURE

1. Measure the mean diameter (d) of the wire of the spring, mean diameter of the spring (D) with a vernier caliper or a micrometer by taking readings at three different places along the length of the specimen. Also note down the number of coils (n) and height of the spring (h).
2. Place the spring on hooks fixed to the upper and middle cross heads of the spring testing machine. Tighten the nuts on the top hook so that the spring is properly held in position.
3. Close the release valve of the loading unit. Apply loading by hand operation at constant rate till the load is reached about 100 kg and then release the load to zero by opening the release valve. Continue this process for three times to ensure that the spring behaves elastically.
4. Set the load pointer and deformation pointer either to zero or to a convenient value.
5. Close the release valve and apply load on the specimen without any jerk manually. At regular increments of applied load, observe the total elongation of the spring. Continue the measurement till 5 to 6 observations are made and then release the load to zero.
6. Plot the load- deflection graph from the calculated values and determine, the ratio $W/\boldsymbol{\delta}$ by determining the slope of the initial straight portion of the curve.
7. Compare the results obtained with the standard values and write inference.

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of vernier caliper	
2	Least count of loading unit	
3	Number of active coils in the spring,n	
4	Height of spring, h (mm)	

Diameter of the spring wire (d)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter d = MSR+(VSR x LC) (mm)	Mean diameter of the spring wire (d) mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

Outer diameter of the spring (OD)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter OD = MSR+(VSR x LC) (mm)	Mean Outer diameter of the spring (OD)
1	Trial 1				
2	Trial 2				
3	Trial 3				

Load – deflection observation table

SL NO	Load W_i (kg)	Load W (N)	Scale reading (mm)		Spring deformation δ= (Final – Initial) (mm)
			Initial	Final	
1					
2					
3					
4					
5					
6					
7					
8					

(INSERT GRAPH)

SAMPLE CALCULATION

Serial number of the observation –

Applied load W = W_I x 9.81=

Spring deformation, **δ** = (Final scale reading – Initial scale reading)

=

CALCULATIONS

Mean diameter of the spring, D = OD – d =

Pitch of the coil, s = $\frac{h}{n}$ =

Angle of helix, α (degree) = $\tan^{-1}[\frac{s}{\pi D}]$ =

Spring index = D/d =

Stiffness of spring (from graph), k = $\frac{W}{\delta}$ =

Modulus of elasticity of the material of the spring, E= 200000 MPa (Assumed)

Modulus of rigidity, G can be calculated from the equation for deflection

$$\boldsymbol{\delta} = [\frac{1}{d^4} \text{ x } (8WD^3 n \sec\alpha)] \text{ x } [(\frac{1}{G} \text{ x } \cos^2\alpha) + (\frac{1}{E} \text{ x } 2 \sin^2\alpha)]$$

$$(\frac{1}{G} \text{ x } \cos^2\alpha) = [\frac{d^4}{8n\sec\alpha\, D^3 \frac{W}{\delta}}] - (\frac{1}{E} \text{ x } 2 \sin^2\alpha)]$$

Corresponding to a force of …………………..N

a) Maximum shear stress due to twisting moment, $\tau = \frac{1}{\pi d^3}$ x (8WD cosα)

b) Maximum bending stress due to bending moment, $\sigma = \frac{1}{\pi d^3}$ x (16WD sinα)

c) Maximum shear stress in the plane of wire, $\tau_{max} = \frac{1}{\pi d^3}$ x (8WD)

RESULTS

SL NO	PARTICULARS	VALUE
1	Spring index	
2	Stiffness of spring, k	
3	Modulus of rigidity, G	
4	Corresponding to a force of …………………..N	
	a) Maximum shear stress due to twisting moment, τ	
	b) Maximum bending stress due to bending moment, σ	
	c) Maximum shear stress in the plane of wire, τ_{max}	

INFERENCE

Experiment no: Date:

5. DOUBLE SHEAR TEST ON MILD STEEL ROD

AIM

To determine the shear strength of the material of the test specimen.

PRINCIPLE

A type of force which causes or tends to cause two contiguous parts of the body to slide relative to each other in a direction parallel to their plane of contact is called the shear force. With such a shear force, the plane of contact will be subjected to a state of pure shear stress. The stress required to produce a fracture in the plane of cross-section is called shear strength. The method of determing the strength consists of subjecting a suitable length of specimen in full cross section to double shear using a suitable test rig. If "A" is the cross sectional area of the specimen and F_u is the maximum shear force applied on the double shear test specimen, then,

Shear strength, $\tau_{max} = \frac{F_u}{2A}$

APPARATUS:

1. Compression testing machine or universal testing machine
2. Specimen
3. Shearing attachment
4. Vernier caliper or a micrometer
5. Weighing balance

DIAGRAM:

DOUBLE SHEAR TEST SET UP

PROCEDURE

1. Measure the mean diameter (d) of the specimen with a vernier caliper or a micrometer by taking readings at three different places along the length of the specimen and calculate the mean area of cross section (A).
2. Calculate the approximate maximum load on the specimen by assuming shear strength of 400 MPa and accordingly set the loading range of the machine.
3. Note down the mass of the plunger (P_m) of the test rig.
4. Select appropriate shear shackles based on the specimen diameter and insert them in to their appropriate slots, one on the plunger and two on the supporting frame.
5. Insert the plunger into its slot and simultaneously insert the test specimen through the shackles.
6. Place the entire assembly on the platform of the compression testing machine or universal testing machine and bring the top plate in contact with the top of plunger.
7. Close the return valve of the loading unit. Switch on the machine and apply load on the specimen without any jerk by gradually opening the control valve. Set the loading rate in such a way that the specimen deforms at about 10 mm per minute.
8. Continue the loading till the specimen fails and note down the failure load (Pu).
9. Compare the results obtained with the standard values and write inference.

OBSERVATION AND CALCULATIONS

Mean diameter of rod (d)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter = MSR+(VSR x LC) (mm)	Mean diameter (d) mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of vernier caliper	
2	Range of loading unit	
3	Least count of loading unit	
4	Least count of weighing balance	
5	Mass of plunger, P_m	
6	Failure load, P_u	

CALCULATIONS

Mean area of cross section of the rod, A = $\frac{\pi \times d^2}{4}$ =

Approximate ultimate load on the specimen = 400 A =

Total maximum shear force on the specimen $F_u = 1000\ P_u + 9.81\ P_m =$

Shear strength, $\tau_{max} = \frac{F_u}{2A}$ =

RESULTS

The shear strength of the material of the test specimen=

INFERENCE

Experiment no: **Date:**

6. HARDNESS TEST

(a) BRINELL HARDNESS TEST

AIM

To determine the hardness of the given specimen using Brinell hardness test.

PRINCIPLE

Indentation hardness is the number related to the area or to the depth of the impression made by an indenter of fixed geometry under a known static load. The method of determination of indention hardness number is non-destructive and hence this is widely used for quality control of products made of ductile materials. The principle used in finding the brinell hardness number (HB) is based on the resistance of the material to permanent indentation. If the surface of the material is indented with a hardened ball, with a force, then the brinell hardness number is defined as the ratio of the applied force to the spherical area of the indentation. Even though the unit of this value is the same as that of stress, the value alone is expressed as a whole number.

The brinell hardness number varies with the applied load, F and the ball diameter, D. Hence Indian Standards Institution has standardized a factor F/D^2 and it is essential to specify the value of F/D^2 selected along with HB. For a correct assessment of brinell hardness number, the ratio of the diameter of the indentation (d) to the diameter of the ball (D) must lie between 0.30 and 0.50 and in no case it should be more than 0.60. Further, the nature of plastic flow under load on thin specimen is appreciably different from that of thick specimen. Hence the thickness of test specimen must be at least 10 times the depth of the indentation and never be less than 8 times the depth of the indentation.

If F is the applied load in kg, D is the ball diameter in mm and d is the diameter of indentation in mm, then the brinell hardness number is given by

$$\mathrm{HB} = \frac{2F}{\pi D(D-\sqrt{D^2-d^2})}$$

Relation between Brinell hardness and Rockwell hardness (with 1.5875 mm dia. steel ball and a total load of 100Kg for scale B) is given by $1/HB = (130-HR_n)/7300$

In the case of steel, Relation between Brinell hardness and tensile strength of steel is given by Tensile strength of the steel in $kg/cm^2 = 35$ HB

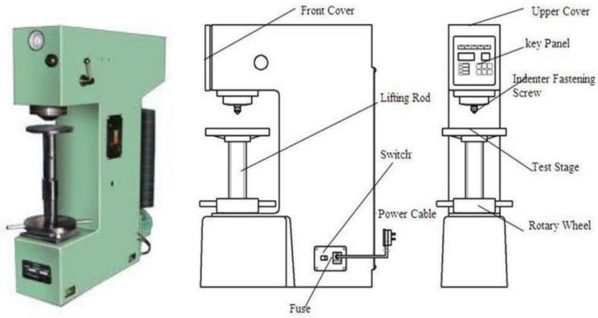
Following table gives the specification for the brinell hardness test.

Material	HB (kN/mm^2)	Specimen thickness (mm)	F/D^2	Ball diameter (D) mm	Load F (kg)	Duration of load application (s)	Typical test material
Ferrous metals	450 to 140	>6	30	10		10	Case hardened steel
		6 to 3		5	750		
		< 3		2.5	187.5		
	Less than 140	>6	30	10	3000	30	Steel, cast iron
		6 to 3		5	750		
		< 3		2.5	187.5		
Non Ferrous metals	130 to 35	>6	10	10	1000	30	Copper, Brass, Aluminium
		6 to 3		5	250		
		< 3		2.5	62.5		
	35 to 8	>6	2.5	10	250	60	Lead, Tin and their alloys
		6 to 3		5	62.5		
		< 3		2.5	15.625		

APPARATUS:

1. Brinell hardness testing machine,
2. Aluminum, steel, brass specimens
3. Ball indenter
4. Optical microscope(Brinell microscope)

DIAGRAM:



BRINELL HARDNESS TESTING MACHINE

PROCEDURE

1. Ensure that the surface of the test specimen is clean by oil, grease, dust etc.
2. Assess the type of material and depending on the thickness of the specimen, select appropriate diameter of the indenter. Insert the indenter to the ball holder of machine.
3. Based on the diameter of indenter selected and type of material, select the load to be applied the specimen. Place the corresponding counter weights on the suspended rod.
4. Keep the hand lever at the "load position" (marked B) and start the machine.
5. Turn the hand lever to the "unload position"(marked A). The suspended load along with the weights will be lifted up and wait till the weight hanger reaches its top position.
6. Place the test specimen on the testing table. Turn the hand wheel till the surface of the test specimen touches the ball indenter. Ensure that the distance between the centre of indentation and edge of the specimen is not less than 2.5 times the diameter of the diameter of the indentation and the distance between centres of two indentations is not less than 4 times the diameter of the indentation.
7. Continue turning the hand wheel till the small pointer reaches the red dot. This corresponds to a minor load of 10 kg applied on the specimen. Check whether the long pointer is at zero and if required, adjust the outer ring of the dial to bring the zero to the long pointer.
8. Turn the hand lever 10 the "load position"(marked B). Once the long pointer of the dial reaches a steady position, Start a stopwatch and wait till the specified time period is reached. At the end point of time period, turn the hand lever to the "unload

position"(marked A). The long pointer will turn back and will come to a rest. The corresponding value on the dial indicates the brinell hardness number (HB) (this corresponds to the depth of indentation) and note down the value.

9. Turn back the hand test wheel. Identify the indentation with a chalk mark and remove the specimen from the test platform.
10. Turn the hand lever to the "load position"(marked B) and switch off the machine.
11. Measure the diameter of impression on two perpendicular axis with the brinell microscope. Calculate the brinell hardness number using the formula. The hardness must be provided along with the details of the diameter of the ball used, applied load and time duration. For example HB 5/750/15 =50 denotes that, the test was conducted using a steel indenter of 5mm diameter under a test load of 750 kg applied for a duration of 15 seconds, the brinell hardness number was 50.

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of indicator dial guage	
2	Least count of Brinell microscope	
3	Equivalent mass of each loose weights	250kg
4	Equivalent mass of weight hanger	250kg

SL NO	Material of the specimen	Diameter of indenter (D) (mm)	F/D^2	Test load F (kg)	Duration of load application (s)
1					
2					
3					

SL NO	Material of the specimen	HB (from dial)	Diameter of indentation(mm)			HB (by calculation)
			d_1	d_2	Mean	

SAMPLE CALCULATION

Serial number of the observation –

Mean diameter of indentation, d = $\frac{d_1+d_2}{2}$ =

CALCULATIONS

Steel

HB………../…………/………. $=\frac{2F}{\pi D(D-\sqrt{D^2-d^2})}$ =

Brass

HB………../…………./………. $=\frac{2F}{\pi D(D-\sqrt{D^2-d^2})}$ =

Aluminium

HB………../…………../………… $=\frac{2F}{\pi D(D-\sqrt{D^2-d^2})}$ =

Rockwell hardness number, RHB= (130 HB-7300)/HB=

Tensile strength of the steel = 35 HB kg/cm^2 = 3.5 HB MPa

RESULTS

SL NO	PARTICULARS	VALUE
1	HB of steel	
2	HB of brass	
3	HB of aluminium	
4	Tensile strength of the steel	

INFERENCE

(b) ROCKWELL HARDNESS TEST

AIM

To determine the hardness of the given Specimen using Rockwell hardness test.

PRINCIPLE

Indentation hardness is a number related to the area or to the depth of the impression made by an indenter of fixed geometry under a known static load. The method of determination of indentation of products hardness made of ductile number materials is non-destructive and hence this is widely used for quality control. The principle used in finding the Rockwell hardness number (HR) is based on the concept that the depth of penetration of an indenter is inversely proportional to the hardness. This method is used for testing of hardness over a wide range of material harnesses. Depending on the types of material, a ball type and cone type indenters are used. Depending on the type indenter used and the load applied, there can be, in general, three scales for reading the Rockwell hardness (Scale A, Scale B and Scale C).

For soft materials, a 1.5875 mm (1/16 inch) diameter steel ball indenter is to be used and readings are to be observed from scale B. Accordingly, the hardness is represented as HR_B. Similarly, for hard materials, 120° diamond cone indenter and scale C is to be used. The corresponding Rockwell hardness number is represented as RH_c. For hardest materials, same cone is used, but with scale A and the hardness value is represented as HR_A. However, depending on the scale to be selected, the load to be applied varies.

If h is the depth of indentation in units of 0.002 mm and K is an index the value of which depends on the type of indenter used (for diamond cone, K = 100 and for steel ball K= 130) then the Rock well hardness is given by

HR = K-h/0.002

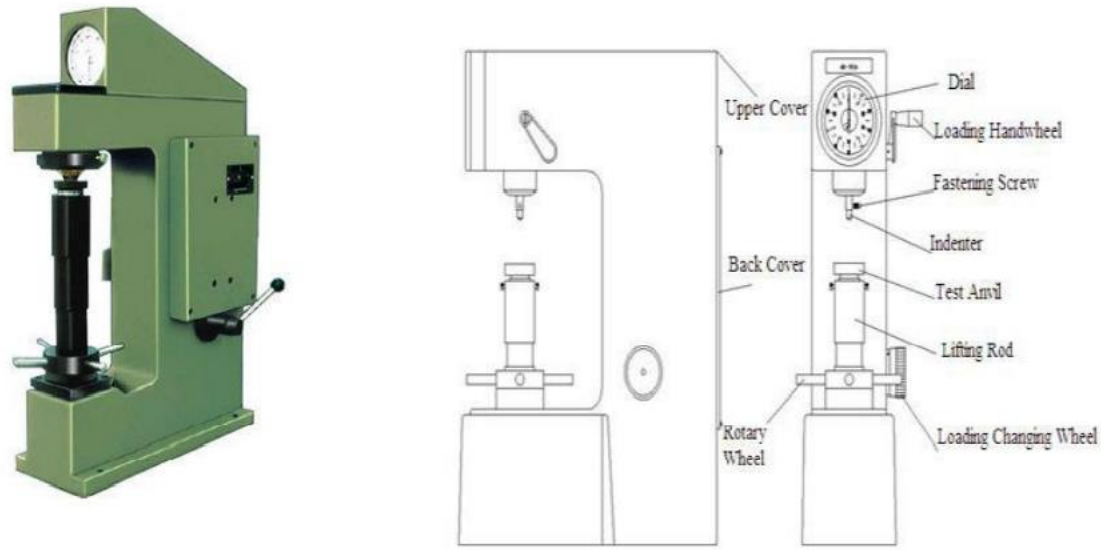
Following table gives the specification for the Rockwell hardness test (for B and C scales)

Rockwell Hardness Number	Indenter type	Load (kg)			Symbol	Permissible scale range	Typical test material
		Minor	Major	Total			
B	Steel ball	10	90	100	HR_B	25 to 100	Non-ferrous metal, annealed steel
C	Diamond cone	10	140	150	HR_c	20 to 67	Annealed case hardened steel

APPARATUS:

1. Rockwell hardness testing machine
2. Specimen for testing
3. Black diamond cone indenter

DIAGRAM:



ROCKWELL HARDNESS TESTING MACHINE

PROCEDURE

1. Ensure that the surface of the test specimen is smooth and is free from oil and dirt.
2. Depending on the type of material, select the type of indenter and attach it to the slot of the machine
3. Select appropriate load to be applied by turning the "load selector disc". Keep the load lever at position A (unload position).
4. Check whether the long pointers at 50 in "C scale" when the small pointers at 10. If not, by rotating the outer ring of the dial, bring the reading 50 of the C scale to the position of the pointer.
5. Place the test specimen on the testing table. Turn the hand wheel till the surface of the test specimen touches the indenter. Ensure that the distance between the centre of indentation and edge of the specimen as well as that between the centres of indentations is not less than 2.5 mm for diamond indenter and 4 mm for ball indenter.
6. Continue turning the hand wheel till the long pointer rotates 2 % times and stops at "zero". Turn the hand wheel further till the small pointer reaches the red dot (value 3). This corresponds to a minor load of l10 kg applied on the specimen.
7. Turn the load lever to the position B (load position). Once the long pointer of the dial reaches a steady position, turn the load lever back to the position A. Turning the lever back from B to A must be done slowly and without any jerk.
8. The long pointer will turn back and will come to a rest. The corresponding value on the dial (B or C, as the case may be) indicates directly the Rockwell hardness number.

9. Turn back the hand wheel and remove the specimen from the test platform.
10. Repeat steps 5 to 9 five times and report the average value, rounded off to the nearest whole number as the Rockwell hardness number

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Range of B scale	
2	Range of C scale	

SL NO	Material of the specimen	Hardness scale	Total load (kg)	Rockwell hardness number			Mean HR
				HR_1	HR_2	HR_3	

SAMPLE CALCULATION

Serial number of the observation –

Rockwell hardness number, HR $= \frac{1}{3}(HR_1 + HR_2 + HR_3) =$

RESULTS

HR_C of Steel =

INFERENCE

Experiment no: **Date:**

7. CHARPY IMPACT TEST

AIM

To determine the impact strength of the material using Charpy test specimen.

PRINCIPLE

Impact test is conducted to determine the behaviour of materials when subjected to high rate of loading (impact). It measures the energy absorbed in breaking the specimen by a single blow or impact. In Charpy impact test, a notched specimen having standardized dimensions is supported at both end as a simple beam and broken by a falling pendulum on the face opposite to and immediately behind the notch. The energy absorbed, as determined by the subsequent rise of pendulum, is a measure of the impact strength.

The test specimen must be machined all over and 55 mm long and of square cross section with 10 mm sides. The notch is made at the centre of the specimen. There can be three types of notches namely, V-notch (IS 1757-1988), U-notch (IS 1499-1977) and Key-hole notch and the dimensions of the notch varies with the type. The V type notch should have an internal angle of $45^o \pm 2^o$ and a depth of 2 ± 0.11 mm.

If "A" is the area of cross section of the specimen under the notch and "K" is the average energy absorbed by the test specimen, then

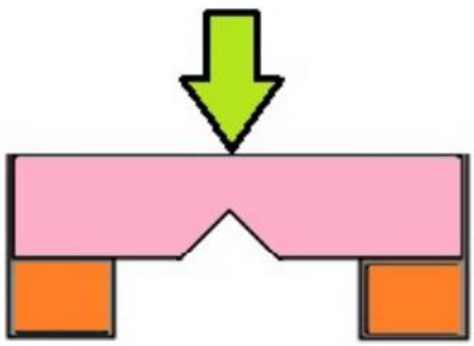
Notch impact strength, I =K/A (J/cm^2)

APPARATUS:

1. Impact test machine
2. Test specimen of mild steel
3. Charpy setting gauge
4. Vernier caliper

DIAGRAM:

IMPACT TESTING MACHINE



SPECIMEN SET UP

PROCEDURE

1. Fix the latching tube to the locating pins for the Charpy test (140° with respect to the vertical). Lift the pendulum and keep it safely at a height with the help of special hook suspended from the latching tube. Fix the Charpy striker to the hammer.
2. Inspect the specimen and the notch for any irregularity. Measure the dimensions of the specimen and the depth of specimen at the notch.
3. Place the test specimen on the "Charpy Izod block" in such a way that the pendulum will strike on the face opposite to the notch.
4. Align the centre of the notch with respect to the centre of supports by means of Charpy setting gauge and firmly fitted to the bearing housing at the side of the columns.
5. Bring down the hammer slowly and touch the striker to the test specimen. Adjust the indicating pointer to a value of 300 J.
6. Adjust the pointer carrier in such a way that it just touches the indicating pointer.
7. Lift the pendulum till it gets latched to the latching tube
8. Release the latch by operating the handle and there by allow the pendulum to swing freely and break the specimen.
9. After the rupture of the specimen, stop swinging of the pendulum by operating the brake.
10. Read the energy absorbed by the specimen directly from the dial as indicated by the pointer.
11. Remove the broken pieces of the test specimen and observe the fracture surface.

OBSERVATION AND CALCULATIONS

SL NO	PARTICULARS	VALUE
1	Range of Charpy Scale	
2	Least count of Charpy Scale	
3	Least count of vernier caliper	
4	Type of cross section of specimen	
5	Type of notch	V-type
6	Initial energy stored in the hammer	300 J
7	Angle of drop	140^o
8	Effective weight of pendulum	21.3 kg
9	Velocity of striking	5.308 m/s
10	Size of specimen – breadth x width x length (mm)	

SL NO	PARTICULARS	MSR (mm)	VSR	Value = MSR+(VSR x LC) (mm)
1	Width of the specimen at notch , a			
2	Depth of the specimen at notch , d_1			
3	Depth of the specimen at notch , d_2			

CALCULATIONS

Average depth of specimen at notch, d = $\frac{1}{2}(d_1 + d_2)$ =

Area of cross section below the notch, A = $\frac{1}{100}$ (a x d) =

Energy absorbed, K_1 =

Error value of Charpy Scale =

Corrected energy absorbed, K = Energy absorbed – error value of Charpy Scale

=

Notch Impact strength, I = $\frac{K}{A}$ =

RESULTS

Energy absorbed by the Izod test specimen =

Notch Impact strength of Izod test specimen of square cross section with V notch =

INFERENCE

Experiment no: **Date:**

8. IZOD IMPACT TEST

AIM

To determine the impact strength of the material using Izod test specimen.

PRINCIPLE

Impact test is conducted to determine the behaviour of materials when subjected to high rate of loading (impact). It measures the energy absorbed in breaking the specimen by a single blow or impact. In Izod impact test, a notched specimen having standardized dimensions is supported vertically on one end as a cantilever beam and broken by a falling pendulum on the face and at the tip of the cantilever projection. The energy absorbed, as determined by the subsequent rise of pendulum, is a measure of the impact strength.

The test specimen must be machined all over and depending on the number of notches made, the total length of specimen varies. The square cross section of the specimen is to have 10 mm sides. The V type notch should have an internal angle of $45^{o} \pm 2^{o}$ and a depth of 2 ± 0.11 mm.

If "A" is the area of cross section of the specimen under the notch and "K" is the average energy absorbed by the test specimen, then

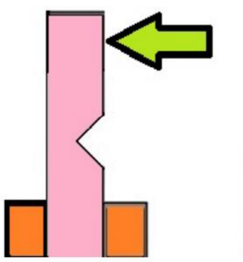
Notch impact strength, I $=$K/A (J/cm^2)

APPARATUS:

1. Impact test machine
2. Test specimen of mild steel
3. Izod setting gauge
4. Vernier caliper

DIAGRAM:

IMPACT TESTING MACHINE



SPECIMEN SET UP

PROCEDURE

1. Fix the latching tube to the locating pins for the Izod test (90° with respect to the vertical). Lift the pendulum and keep it safely at a height with the help of special hook suspended from the latching tube. Fix the Izod striker to the hammer.
2. Inspect the specimen and the notch for any irregularity. Measure the dimensions of the specimen and the depth of specimen at the notch.
3. Place the test specimen on the "Charpy Izod block" in such a way that the pendulum will strike on the face of the notch.
4. Align the centre of the notch with respect to the centre of supports by means of Izod setting gauge and clamp the specimen using clamping screw.
5. Bring down the hammer slowly and touch the striker to the test specimen. Adjust the indicating pointer to a value of 170 J.
6. Adjust the pointer carrier in such a way that it just touches the indicating pointer.
7. Lift the pendulum till it gets latched to the latching tube
8. Release the latch by operating the handle and there by allow the pendulum to swing freely and break the specimen.
9. After the rupture of the specimen, stop swinging of the pendulum by operating the brake.
10. Read the energy absorbed by the specimen directly from the dial as indicated by the pointer.
11. Remove the broken pieces of the test specimen and observe the fracture surface.

OBSERVATION AND CALCULATIONS

SL NO	PARTICULARS	VALUE
1	Range of Izod Scale	
2	Least count of Izod Scale	
3	Least count of vernier caliper	
4	Type of cross section of specimen	
5	Type of notch	V-type
6	Initial energy stored in the hammer	170 J
7	Angle of drop	90^o
8	Effective weight of pendulum	21.3 kg
9	Velocity of striking	3.994 m/s
10	Size of specimen – breadth x width x length (mm)	

SL NO	PARTICULARS	MSR (mm)	VSR	Value = MSR+(VSR x LC) (mm)
1	Width of the specimen at notch , a			
2	Depth of the specimen at notch , d_1			
3	Depth of the specimen at notch , d_2			

CALCULATIONS

Average depth of specimen at notch, d = $\frac{1}{2}$ x (d_1+ d_2) =

Area of cross section below the notch, A = $\frac{1}{100}$ x (a x d) =

Energy absorbed, K_1 =

Error value of Izod Scale =

Corrected energy absorbed, K = Energy absorbed – error value of Izod Scale

Notch Impact strength, I =$\frac{K}{A}$

RESULTS

Energy absorbed by the Izod test specimen =

Notch Impact strength of Izod test specimen of square cross section with V notch =

INFERENCE

Experiment No: Date:

9. PREPARATION OF CONCRETE CUBES, CYLINDERS AND BEAMS

REF:

IS: 516-1959

AIM:

To prepare concrete cubes, cylinder and beam specimens

(A) CUBES:

APPARATUS:

1. Cube Mould (150x150x150 mm)

2. Tamping bar (16 mm diameter and bull-nosed)
3. Steel Float/Trowel

PROCEDURE:

1. Clean the cube-mould properly and apply oil on inner surface of mould. But no oil should be visible on surface.
2. Fix the cube mould with base plate tightly. No gap should be left in joints so that cement-slurry doesn't penetrate.
3. Place the mould on levelled surface.
4. Place concrete into mould in three layers. Compact each layer by giving 35 blows of tamping bar.
5. Remove excess concrete from the top of mould and finish concrete surface with trowel. Make the top surface of concrete cube even and smooth.
6. Leave the mould completely undisturbed for first four hours after casting.
7. After ending undisturbed period, put down casting date and item name on the top of concrete specimen with permanent marker.
8. Remove the cube specimens from mould after 24±½ hours of casting. For removing specimen from mould, first loosen all nut-bolts and carefully remove specimen because concrete is still weak and can be broken.

9. Immediately after removing, put the specimen into a tank of clean water for curing. Make sure cube specimen is fully submerged in water.
10. After 28 days of curing take out specimens from water tank and send to laboratory for testing.

(B) CYLINDERS

APPARATUS:

1. The cylinder mould (150mm diameter and 300mm height)
2. Tamping Bar

CYLINDER MOULD   TAMPING BAR

PROCEDURE:

1. The test specimen should be made as soon as practicable after the concrete is filled into the mould in layers approximately 5 cm deep. Each layer is compacted either by hand or by vibration.
2. When compacted by hand, the standard tamping bar is used and the stroke of the bar should be distributed in a uniform manner. The number of strokes for each layer should not less than 30.
3. The stroke should penetrate in to the underlying layer and the bottom layer should be rodded throughout its depth.
4. After top layer has been compacted, the surface of the concrete should be finished level with the top of the mould, using a trowel and covered with a glass or metal plate to prevent evaporation.
5. The test specimen should be stored in a place at a temperature of $27° \pm 2°C$ for 24 $\pm$ 0.5 hrs from the time addition of water to the dry ingredients.
6. After this period the specimen should be marked and removed from the moulds and immediately submerged in clean fresh water or saturated lime solution and kept there until taken out just prior to the test. The water or solution in which the specimens are kept should be renewed every seven days and should be maintained at a temperature of $27° \pm 2°c$.

RESULT:

Experiment no: Date:

10. COMPRESSION TEST ON CONCRETE CUBES

AIM:

To determine the compressive strength of concrete cubes.

PRINCIPLE:

Out of many test applied to the concrete, this is the utmost important which gives an idea about all the characteristics of concrete. Compressive strength is the capacity of a material or structure to withstand loads tending to reduce size, as opposed to tensile strength, which withstands loads tending to elongate. In other words compressive strength can be defined as the maximum compressive stress that, under a gradually applied load, a given solid material can sustain without fracture. Compressive strength is calculated by dividing the maximum load by the original cross-sectional area of a specimen in a compression test. Compressive strength of concrete depends on many factors such as water-cement ratio, cement strength, quality of concrete material, quality control during production of concrete.

APPARATUS:

Compression testing machine, concrete cubes

COMPRESSION TESTING MACHINE

PROCEDURE:

1. Remove the specimen from water after specified curing time and wipe out excess water from the surface.
2. Take the dimension of the specimen to the nearest 0.2m
3. Clean the bearing surface of the testing machine

4. Place the specimen in the machine in such a manner that the load shall be applied to the opposite sides of the cube cast.
5. Align the specimen centrally on the base plate of the machine.
6. The movable portion should be adjusted such that it touches the top surface of the specimen.
7. Apply the load gradually without shock and continuously at the rate of 14KN/cm^2/minute till the specimen fails.
8. Record the maximum load and note any unusual features in the type of failure.

OBSERVATIONS AND CALCULATIONS:

1. Date of casting =
2. Date of testing =
3. Size of the cube =
4. Area of the specimen (calculated from the mean size of the specimen) =

Sample no	Weight of cube (kg)	Area of the specimen(mm^2)	Load P (kN)	Compressive strength(N/mm^2)	Average (N/mm^2)
1					
2					
3					

SAMPLE CALCULATION

Serial number of the observation –

Area of the specimen =

Load =

Compressive strength $= \frac{1000 \times P}{A}$

RESULT:

Compressive strength of the concrete cube at 28 days =

INFERENCE:

Experiment no: **Date:**

11. SPLIT TENSILE TEST ON CONCRETE CYLINDERS

AIM:

To determine the split tensile strength of concrete.

PRINCIPLE :

The tensile strength of concrete is one of the basic and important properties. Split tensile strength test on concrete cylinder is a method to determine the tensile strength of concrete. The concrete is very weak in tension due to its brittle nature and is not expected to resist the direct tension. The concrete develops cracks when subjected to tensile forces. Thus, it is necessary to determine the tensile strength of concrete to determine the load at which the concrete members may crack.

APPARATUS:

1. Compression testing machine
2. Two packing strips of plywood 30 cm long and 12mm wide.

COMPRESSION TESTNG MACHINE

PROCEDURE:

1. Take the wet specimen from water after 28 days of curing
2. Wipe out water from the surface of specimen
3. Draw diametrical lines on the two ends of the specimen to ensure that they are on the same axial place.
4. Note the weight and dimension of the specimen.
5. Set the compression testing machine for the required range.
6. Keep are plywood strip on the lower plate and place the specimen.
7. Align the specimen so that the lines marked on the ends are vertical and centred over the bottom plate.

8. Place the other plywood strip above the specimen.
9. Bring down the upper plate to touch the plywood strip.
10. The load shall be applied without shock and increased continuously at a nominal rate within the range 1.2 N/ (mm^2/min) to 2.4 N/ (mm^2/min).
11. Note down the breaking load(P)

CALCULATIONS:

1. Date of casting =
2. Date of testing =
3. Weight of cylinder =
4. Diameter of the cylinder (D) =
5. Length of the cylinder (L) =
6. Failure load (P) =
7. Tsp $=\frac{2 \times P \times 1000}{\pi D L}$

RESULT:

Split tensile strength of given concrete at 28 days =

INFERENCE:

Experiment no: **Date:**

12. BENDING TEST ON WOOD

AIM

To observe the behaviour of wood under bending and to calculate its modulus of elasticity as well as the modulus of rupture of wood.

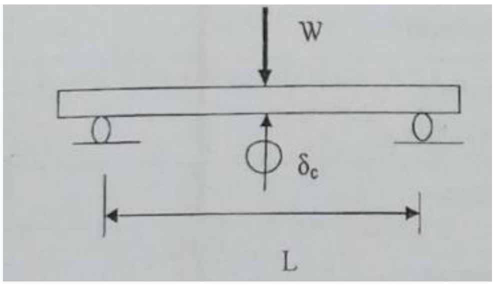
PRINCIPLE

Wooden beams are widely used for various purposes. The factors that affect the strength and behaviour of wooden beams include slope and closeness of grain, density, moisture content various defects in timber (knot, rind galls, shakes, etc.), etc. So, a bending test in a sample wooden specimen only will help in obtaining information required for the design of timber beams such as modulus of elasticity and modulus of rupture._When a beam is under pure bending, the general equation for bending $\frac{M}{I} = \frac{f}{y} = \frac{E}{R}$

where M = Bending moment, I = moment of inertia of the section, f = bending stress developed, y = distance of any fibre measured from the neutral axis, E = modulus of elasticity

R = radius of curvature of the beam

can be used to determine the bending stress developed across the depth of any section. In laboratories, two point loading system is generally adopted to achieve a pure bending condition on a beam specimen. IS 1708 (I 969) specifies 50 mm x 50 mm x 750 mm size standard beam specimen and standard third point loading for bending test on wood.
Consider a beam of span L subjected to a load of W applied at its mid-point.



Now, from the bending formula, we can write

Modulus of rupture, $f_r = M_r\, y / I$ where, $M_r = \frac{1}{4} W_r L$; $y = \frac{1}{2} d_m$; $I = \frac{1}{12} (b_m \times d_m^3)$

From the general expression for deflection, we can write

Deflection at centre, $\boldsymbol{\delta}c = \frac{1}{48EI} \times W L^3$

APPARATUS:

1. Wood testing machine
2. Standard specimen of wood
3. Proving ring

4. Dial gauge
5. Graph paper
6. Steel rule
7. Vernier caliper or a micrometer

DIAGRAM:

WOOD TESTING MACHINE

PROCEDURE

8. Measure the mean depth (d_m) and mean breadth (b_m) of the specimen with a vernier caliper or a micrometer by taking readings at three different places along the length of the specimen.
9. Mark the mid-point and support points on the specimen.
10. Fix the proving ring to the top cross head of the wood testing machine and bring it in the contact to the specimen through a roller placed at mid span.
11. Set the loading rate to about 2.5 mm per minute.
12. Place a dial gauge in such a way that the mid span deflection of the specimen be measured. Set dial gauge reading to zero.
13. Switch on the machine and apply load on the specimen without any jerk.
14. At regular interval of proving ring dial reading, observe the deflection of the beam at mid-point. Continue the measurement till 5 to 6 observations and remove the dial gauge from the test set up.
15. Continue the loading till the first facture is observed on the bottom surface of the specimen is within the constant moment zone. Note down the corresponding load (W_r) and continue till the specimen fails completely.
16. Remove the specimen, observe the fracture pattern and sketch the position and pattern of the facture.
17. Plot the load – deflection graph from the calculated values and determine, the ratio W/ $\boldsymbol{\delta}$ by determining the slope of the initial straight portion of the curve.
18. Compare the results obtained with the standard values and write inference.

OBSERVATION AND CALCULATIONS

Least count of measuring instruments

SL NO	PARTICULARS	VALUE
1	Least count of vernier caliper	
2	Least count of dial gauge	
3	Least count of proving ring (proving ring constant)	
4	Rate of loading (mm/minute)	
5	Span of beam L (mm)	

Mean depth of specimen (d_m)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter = MSR+(VSR x LC) (mm)	Mean depth (d_m) mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

Mean breadth of specimen (b_m)

SL NO	PARTICULARS	MSR (mm)	VSR	Diameter = MSR+(VSR x LC) (mm)	Mean breadth (b_m) mm
1	Trial 1				
2	Trial 2				
3	Trial 3				

Load – deflection observation table

SL NO	Proving ring reading		Deflection at mid span, δ_C (mm)	
	Dial gauge reading	Load (kN)	Dial gauge reading	δ_C (mm)

SAMPLE CALCULATION

Serial number of the observation –

Load = (Dial gauge reading) **x** (proving ring constant)

=

Deflection at mid span, δ_C = (Dial gauge reading) **x** (Least count of dial gauge)

=

(INSERT GRAPH)

CALCULATIONS

Load corresponding to the first facture of the specimen (W_r) =

Moment of inertia, $I = \frac{1}{12} (b_m \times d_m^3) =$

Moment corresponding to the fractured load $M_r = \frac{1}{4 \times 1000}$ x W_r L=

Distance of any fibre measured from the neutral axis, y = $\frac{1}{2}$ d_m =

Modulus of rupture, $f_r = \frac{1}{I}$ x (M_r y) x 10^6 =

Stiffness of beam (slope of curve), $\frac{W}{\delta}$=

Young's modulus = $\frac{1}{48I}$ x $\frac{W}{\delta}$ x L^3 x 10^3=

RESULTS

Modulus of rupture of wood =

Stiffness of the beam =

Young's modulus of the material of the wood =

INFERENCE

Experiment no: **Date:**

13. MODULUS OF ELASTICITY OF CONCRETE

AIM :

To determine the modulus of elasticity of concrete.

OVERVIEW:

The stress – strain curve of the concrete is nonlinear both in compression and tension. Hence the modulus of elasticity of concrete gives the rate of stress to strain and varies depending on the point selected on the stress strain curve. Two moduli of elasticity that is tangent modulus and secant modulus are often defined for concrete. Tangent modulus at any point on stress strain curve is defined as the slope of line drawn tangential to the stress strain curve at that point and it gives the instantaneous modulus at the given state of stress/strain. Secant modulus at any point is defined as the slope of line drawn between the origin and given point on the stress strain curve and it shows the average modulus between the free condition and the given state of stress. Generally for design purpose secant modulus corresponding to a strain of 0.2 percent is taken as the value for modulus of elasticity of concrete. Concrete exhibits considerable amount of creep deformation that is deformation under sustained loading. To eliminate creep and also to achieve seating of gauges, at least two cycles of preloading to the maximum stress is recommended. The stress curve on the third or fourth loading exhibits only a small curvature. The modulus of elasticity increases approximately with the square root of the characteristic compressive strength of concrete. The modulus of elasticity is a function of the modulus of elasticity of the aggregates and the cement matrix and their relative proportions. The modulus of elasticity of concrete is relatively constant at low stress levels but starts decreasing at higher stress levels as matrix cracking develop.

APPARATUS:

a) Compressometer
b) Universal testing machine
c) Concrete cylinder of 15 cm diameter and 30cm long

UNIVERSAL TESTING MACHINE COMPRESSOMETER

PROCEDURE:

A. Setting up of compressometer

1. Take a standard concrete cylinder on the 28 th day after casting from the curing tank. Wipe off the surface.
2. Measure the dimensions and note the weight of specimen.
3. Assemble the top and bottom frame by keeping the spacers in position.
4. Keep the pivot rod on the screws and lock them in position.
5. Keep the tightening screws of the bottom and top frame unscrewed (but not completely).
6. Place the specimen on a level surface.
7. Keep the compressometer centrally on the specimen so that the tightening screw of the bottom and top frame are at equal distance from the two ends.
8. Tighten the screws so that the compressometer is held on the specimen.

B. Testing

1. Calculate the range of testing machine to be set by assuming the compressive strength of the cylinder as $0.8xf_{ck}$, where f_{ck} is the characteristic compressive strength after 28 days of curing in N/mm^2.
2. Calculate the maximum and minimum loads to be applied in each cycle of loading.
3. Place the specimen with compressometer on the lower plate of the universal testing machine and centre it. If the specimen is not capped a plywood or a card board piece may be placed on the top face of the specimen.
4. Remove the spacers by unscrewing the spacer screws attached to the compressometer. Initialize the dial gauge of the compressometer to a convenient value (zero).
5. Apply the load gradually and continuously on the specimen at a rate of 14MPa per minute to the calculated maximum load as calculated for the first cycle of loading. Maintain the load for at least one minute and release it gradually to the minimum load calculated. Observe the dial gauge reading of the compressometer.
6. Repeat the step 5 till there is no residual deformation over loading and unloading (normally by third cycle of loading, this condition will be achieved).
7. Once the above condition is achieved, reduce the load to zero and set the dial gauge of the compressometer to zero.
8. Increase the load gradually at a rate of 14MPa per minute and note the compressometer reading at regular intervals of loads.
9. Once the test is over, take out the specimen from UTM. Place back the spacers to the compressometer and remove the compressometer from the specimen.
10. A graph is plotted with strain along X-axis and stress along Y- axis. A straight line is drawn from the origin and extended until it touches the curve at a point. Note down the stress and strain at that point. Its slope will give the secant modulus of elasticity.

OBSERVATION AND CALCULATION:

1. Date of casting =
2. Date of testing =
3. Mix ratio =

4. Water- cement ratio =
5. Weight of cylinder =
6. Diameter of cylinder =
7. Cross sectional area A = $\frac{\pi \times D^2}{4}$ =
8. Characteristic compressive strength f_{ck} =
9. Cylinder compressive strength, P_a = 0.8 x f_{ck} =
10. Approximate value of load, F= $\frac{P_a \times A}{1000}$ =
11. Gauge length L_p in mm =
12. Least count of compressometer (L.C)(mm) =

LOADING

SL.NO	PARTICULARS	VALUE
1	**FIRST CYCLE OF LOADING**	
	Maximum stress to be applied, $f_{c1} = \frac{Cube\ strength}{3} + 0.5$ N/mm^2	
	Load corresponding to f_{c1}, $P_1 = \frac{fc_1 \times A}{1000}$(kN)	
	Load corresponding to the minimum stress of 0.15 N/mm^2 $P_2 = \frac{0.15 \times A}{1000}$ (kN)	
2	**SECOND AND THIRD CYCLE OF LOADING**	
	Maximum stress to be applied, $f_{c3} = \frac{Cube\ strength}{3} + 0.15$ N/mm^2	
	Load corresponding to f_{c3},$P_3 = \frac{fc_3 \times A}{1000}$ (kN)	
	Load corresponding to the minimum stress of 0.15 N/mm^2 $P_4 = \frac{0.15 \times A}{1000}$ (kN)	

Sl.No	Load (KN)		Compressometer reading (r) (mm)	Deformation (ΔL) = r x(L.C/2) mm
1	P_0			
	P_1			
	P_2			
2	P_2			
	P_3			
	P_4			
3	P_2			
	P_3			
	P_4			

Sl.No	Load (kN)	Compressometer reading (r)mm	Deformation (ΔL) = r x$(\frac{L.C}{2})$ mm	Stress(N/mm^2) $= \frac{1000 \times load}{Area}$	Strain e = $\frac{\Delta L}{L_p}$
1					
2					
3					
4					
5					
6					
7					
8					
9					

SAMPLE CALCULATION

Serial number of the observation –

Deformation, ΔL = Compressometer reading x $\frac{least\ count\ of\ compressometer}{2}$

$=$

Strain $= \frac{\Delta L}{L_{\mathrm{p}}}$ $=$

Stress $= \frac{1000 \times load}{Area}$ $=$

RESULT:

Modulus of elasticity of given concrete =

INFERENCE: